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**The new Clean Industrial Deal –
How can new EU industrial policy
support sustainability transitions,
competitiveness and resilience?**

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Paula Kivimaa

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The new Clean Industrial Deal – How can new EU industrial policy support sustainability transitions, competitiveness and resilience?

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Abstract

This paper examines the new industrial policy of the European Commission – the Clean Industrial Deal (CID) – based on a historical context of industrial policy development in the European Union (EU) and perspectives from sustainability transitions research. It notes that CID is in essence the EU's reaction to the return of stronger industrial policy more globally since the 2010s and in particular to the EU's decline in competitiveness against China and the United States – as noted in the Draghi report in 2024. CID contains three objectives – linked to competitiveness, sustainability and resilience – with measures that present both synergies and tensions between the three objectives. The paper concludes by arguing that the EU needs a well-targeted and directional industrial policy that leaves openness for innovation and collaboration, but that must direct the development – not only via visions and targets – but gradually strengthening and foreseeable regulatory context that guides the development of innovations, industries and markets. The emergence and scaling of new industries that bring competitiveness and sustainability benefits, needs public sector support via promotion of industrial clusters, investments into education and training, faster permitting and EU-wide industrial subsidies on sustainable industrial production. This also means improving public sector capabilities, which are based on reflexive governance approaches and experimentation.

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1. Introduction

The current European Commission that launched the Competitiveness Compass and Clean Industrial Deal in early 2025, is taking a stronger approach to industrial policy than the previous commission. During the first years of the previous commission, at the turn of the decade, EU industrial policy was argued to be far from effective (Pianta et al., 2020). The recent new policy initiatives are the European Commission's response to the innovation gap and competitiveness problems noted, for instance, in the Draghi report (see Draghi, 2024).

The Clean Industrial Deal embraces three central objectives: boosting competitiveness, mitigating and adapting to climate change, and improving economic resilience. The aim of this paper is to place this policy development in a longer-term historical industrial policy context in the EU and look at it from the perspective of the much-needed zero-carbon socio-technical transitions. I, therefore, use the term 'sustainability transitions' to refer to the pursuit of large-scale systemic socio-technical transitions towards environmental sustainability. Sustainability transitions not only refer to decarbonisation but the need to address other environmental impacts too. Instead of economic resilience, I will address security more broadly, meaning with it different questions around security of supply, security of operation of technical systems, geopolitical security (Kivimaa, 2024) and open strategic autonomy (Kivimaa and Rogge, 2024). The paper will also attempt to discuss synergies and tensions between achieving the triple objective of competitiveness, sustainability transitions and security via industrial policy, and the governance arrangements needed to do so.

The term 'industrial policy' has been used to refer to "structural policies designed to strengthen the efficiency, scale and international competitiveness of domestic industrial sectors" that "typically contains an element of national champion, of self-reliance in bringing about economic growth and development" (Soete, 2007). It has traditionally been described a national endeavour (Di Carlo and and Schmitz, 2023) with variation in characteristics (e.g. visions, policy instruments and governance) and importance across European countries (Johnstone et al., 2021). Modern industrial policy is often associated with innovation (Uyarra et al., 2020), the manufacturing of cleaner technologies such as wind or solar power components (Matsuo and Schmidt, 2019) and the production of non-fossil electricity to power new industries. Green industrial policy mixes have been argued to need complementary measures to traditional industrial policy, such as carbon pricing, regulation, experimentation, measures stabilising climate uncertainty for investors, and a wider group of stakeholder networks (Veugelers et al., 2024).

However, this association with innovation and transitioning to sustainable technologies also involves tensions between the disruptive orientation of innovation and transitions (Kivimaa et al., 2021) and the long-term stability of industrial production and employment. **Hence, industrial policy must simultaneously search for new opportunities while maintaining and transforming old industries at least until the new ones have scaled up. It is also vital that EU industrial policy supports sustainability transitions, while it seeks to improve competitiveness, because the climate and environmental impacts are rapidly intensifying.** As Nilsson et al. puts it: "A major shift is needed, from contemporary industrial policy that mainly protects industry to policy strategies that transform the industry" (Nilsson et al., 2021).

The Ecologic Institute emphasises three policy angles for contemporary EU industrial policy: competitiveness of the overall economy, transformation of incumbent energy-intensive industries, and scaling new key clean technologies (Martini et al., 2024). According to the Draghi report, the EU suffers from a static industrial structure, lack of disruptive innovation and financing, weaknesses of the single market and regulatory barriers to innovation, and has too few top-level academic institutions (Draghi, 2024). Although addressing these problems requires public sector actions on all levels from the EU to the regional, they cannot be purely solved by the public sector. Large established industries hold incumbent power positions and supportive path dependent structures (Kattirtzi et al., 2021; Steen and Weaver, 2017) which may tweak public R&D funding and industrial taxation structures in their favour – rather than to support young and more disruptive entrants. In addition, companies need to not only innovate in terms of costs but also design and performance. Electric vehicles are a good example, where South Korean and Chinese manufacturers are gaining increasing strength, not only in terms of costs but also performance and design. Therefore, disruptive innovation among established companies requires processes of disruption but also of repurposing their skills and assets to new contexts (Kivimaa and Sivonen, 2023; Mäkitie, 2020).

From this setting, this paper explores the background for today's industrial policy, its contemporary challenges, and interconnections to sustainability transitions. It starts with a history of EU industrial policy and its links to sustainability transitions (Section 2). The following section will discuss the Clean Industrial Deal and its possible synergies and tensions from the perspective of different policy objectives (Section 3). The broader governance and policy issues are discussed subsequently (Section 4), before concluding remarks are made (Section 5).

2. History of EU industrial policy and sustainability goals

2.1 From strong industrial policy to market-led approaches

Industrial policy gained popularity in post World War II context, especially in the nation states most affected by the war, and it was initially supported by the establishment of the European Coal and Steel Community in 1952 and Common Agricultural Policy (Soete, 2007). Bianchi and Labory have recognised three phases of post-war industrial policy: (1) strong and selective intervention until the late 1970s; (2) market-led approaches of the 1980s and 1990s; and (3) a pragmatic approach at the turn of century to address the needed deep structural changes with industrial adjustments (Bianchi and Labory, 2020). The current period can be seen as a strong return of industrial policy in many countries across the world.

The first phase was characterised by protective and interventionist policies, identifying and targeting those industries which were expected to create future growth and protecting them with tariffs (Veugelers et al., 2024). Being based on mass production, fossil fuels and energy and resource intensity, the first phase of industrial policy created persistent problems around its environmental effects and rising global inequalities (Schot and Steinmueller, 2018), the former mostly addressed with end-of-pipe solutions (Nilsson et al., 2021) and the latter not properly dealt with. The first phase was found to favour large existing companies, distorting domestic competition and reducing innovativeness (Aghion, 2023). This kind of selective and traditional industrial policy lost popularity from the 1980s onwards in many countries via the rise of neoliberalism and seeing government as inefficient in 'picking winners' (Johnstone et al., 2021).

The second phase saw what is called by some as ‘horizontal industrial policy’ addressing all sectors of the economy and comprising, for example, investments in higher education and research, reforming labour and product markets, and developing new funding mechanisms for innovation (e.g., venture capital and private equity) (Aghion, 2023). In the EU, support for industrial policy shifted from the national level to the European-level in the 1980s, with the European Roundtable of Industrialists established in 1983 and the creation of the Single Market in 1992 to remove internal trade barriers (Soete, 2007). This kind of market-oriented industrial policy was directed to setting supportive frameworks of economic processes and research and innovation to promote industrial activity (Veugelers et al., 2024). While some label this still as industrial policy, others prefer to use the terms ‘technology policy’ or ‘innovation policy’ instead. In effect, industrial policy was not explicitly used in many countries’ policy strategy documents during this phase. Some have described that industrial policy effectively phased out in the EU for a while (Aiginger, 2007; Török, 2007).

European policymaking reduced public industrial policy interventions and pushed back on government involvement in the economy in the 1990s (Pianta et al., 2020). In effect, the Single Market and the Maastricht Treaty have been argued to have reduced policy space for industrial policy, dragging down competitiveness (Pianta et al., 2020). When other countries globally have continued to push stronger industrial policies, the EU lost many of its industries, including, for example, solar, electronics and shipping. Companies, such as Nokia, lost dominance in the early 2000s, leading to substantial job losses and economic decline. This demonstrates the shortcomings of purely enabling policies. Some of the explanations provided also deal with the US having higher RTD support and lowering industrial costs strategy, while European countries’ focus, at the time, on reducing budget deficits and discouraging inflation did not work (Aiginger, 2002). While the water and air pollution caused by industrial production have been of concern since the 1970s, the greenhouse gas emissions were brought to the policy agenda in the 1990s – perhaps playing some role in the decline of industrial policy too.

In summary, the 1990s and early 2000s can be characterised, instead of strong industrial policy, as the rise of innovation policy – although initially the term ‘technology policy’ was used. This relied on new thinking about ‘national innovation systems’ (Lundvall, 2007; Patel and Pavitt, 1994), accompanied by research on regional and sectoral innovation systems. Key points put forward by the innovation systems approach were that institutions and policies are mainly at the national level and that the travel of knowledge outside the socio-cultural milieu in which it is generated is difficult, i.e. knowledge is sticky (Schot and Steinmueller, 2018). This created the basis of identifying and supporting R&D activity on specific industrial clusters, with somewhat less focus on instruments such as tariffs and non-tariff trade barriers.

2.2 Return of industrial policy since 2010s

The third phase of industrial policy can be described by a return or re-emergence of explicit industrial policy during the 2010s. For instance, the European Commission published a new industry-oriented communication in 2012. Similarly, many nation states brought back the industrial policy term in their policy agendas. The British government issued “Industrial Strategy: building a Britain fit for the future” in 2017. Pertaining to sustainability transitions, the US government first introduced antidumping duties¹ for

¹ An anti-dumping duty is a tax set on imported products when they are sold at a price below their fair market value, or “dumped,” in the importing country’s market.

imported solar panels in 2012 and a safeguard tariff² in 2018 (Hufbauer and Jung, 2021). In Germany, there were calls for more active industrial policies in the 2010s too (Aiginger and Rodrik, 2020).

The reinvigoration of industrial policy builds links to innovation, industrial renewal and trade policy. On the one hand, it has been shaped by increasing globalisation and competition for skills (Bianchi and Labory, 2020). On the other hand, it has emphasised the regionality of industrial production. However, the contemporary context, showing more protectionist, nationalist and even autarkical policy approaches by the world's major states, is countering the previous globalisation-oriented industrial policy context.

In the EU, Smart Specialisation at the intersection of industrial and cohesion policy – which can be seen as form of EU-based regional innovation policy – has been actively promoted since 2010. This has partly been a response by the European Commission to a concentration of the EU's manufacturing base in Germany and Central-East economies, while Southern member states have become peripheral (Pianta et al., 2020). However, Smart Specialisation has been argued to be “unlikely to lead to the proclaimed and much-needed economic intra-EU convergence” (Wigger, 2023), partly because it has still continued the horizontal approach of setting framework conditions (Kotnik and Petrin, 2017). It has also failed to encourage innovation in those regions that are most lagging, because the entrepreneurial discovery process is hard to implement in locations of weak networks and technology/skills gaps (Barzotto et al., 2020). Cohesion funding used to support smart specialisation has also suffered from excessive bureaucracy and sometimes corruption (Pianta et al., 2020). Smart specialisation has also been criticised for being based on the traditional science and technology model of innovation and ignoring ecological considerations (Barzotto et al., 2020). However, subsequent work on smart specialisation under the label of Partnership for Regional Innovation during 2021-2023 aimed more explicitly to employ a systemic model of innovation and consider mechanisms supporting sustainability transitions. Overall, it seems that smart specialisation has not yet been an answer to the EU's re-industrialisation and competitiveness efforts.

Ursula von der Leyen's first Commission launched the EU Green Deal. It is an overarching policy package that aims to achieve the goal of climate neutrality by 2050. This increased the tempo and scope of climate action in the EU (Dupont et al., 2024). Soon after its publication, the Green Deal was argued to lack both active industrial policy (industrial policy focus being too narrow) and ambitiousness for the scale of sustainability transitions needed – both aspects suffering from a claimed lack of vision (Pianta and Lucchese, 2020). In addition, the member states were argued to be in differing positions regarding the requirements set by the Green Deal and its implementation package Fit-for-55 (D'Amato et al., 2024). A related development of green industrial policy “represents a transformative agenda that integrates ecological considerations into the heart of industrial planning and decision-making”, which is challenging to create on the pre-existing carbon-intensive industrial base of the EU, despite the Green Deal forming a cornerstone for the shift in industrial policymaking (Veugelers et al., 2024).

Although industrial policy was not visible in the name of the Green Deal, the policy framework contained industrial policy ideas. Economic ideas have shaped climate policy and especially the 'green growth' agenda has dominated global environmental discourse in early 2010s following the market-based climate policy agenda of the 1990s and 2000s

² A safeguard tariff is a temporary import restriction, set by a state to protect its industry from a surge in imports.

(Meckling and Allan, 2020). This also fed into how the EU Green Deal was formulated (da Ponte et al., 2023).

In 2020, the European Commission launched a new industrial strategy for Europe (European Commission, 2020), which was revised already in 2021, impacted by the start of the Covid-19 pandemic. The updated strategy emphasised the need to improve the resilience of the European single market by improving the EU's 'open strategic autonomy' in key areas (Tagliapietra and Veugelers, 2023). The industrial strategy based on the Green Deal led to new legislation, importantly the Critical Raw Materials Act (CRMA) in 2023 and Net-Zero Industries Act (NZIA) in 2024.

NZIA boosts manufacturing capacity in the EU aiming for improved competitiveness and security of supply. It includes measures, such as selecting strategic projects with faster permitting and non-price criteria for public procurement on sustainability, resilience and cyber security. It also contributes to sustainability transitions, by aiming to accelerate cleantech expansion (Kivimaa and Entsaló, 2025).

CRMA has resulted from concerns about the EU's reliance of critical raw materials from external providers, especially China. It stipulates that no more than 65% of critical raw materials should arrive from a single third country and that the EU should provide 10% of its annual consumption for extraction and 40% of annual consumption for processing. It aims at strategic projects to be advanced with funding and faster permitting, monitoring and stress-testing supply chains, risk preparedness obligations on large companies, improving the recovery and recycling of secondary critical raw materials, and diversifying imports. In essence, it is addressing the implications of the energy transition on increasing raw materials provision, but also affected by rising demand of materials by digitalisation and armament manufacturing.

The literature on industrial policy development emphasises some past experiences that can be learned from, such as a lack of European-level public procurement policy, the importance of bilateral trade agreements for industrial developments (Soete, 2007), and a non-optimal institutional framework with unclear objectives and limited resources (Pianta et al., 2020). Further, Pianta et al. suggest some ways forward, such as the importance of having EU-level industrial policy but one that allows also local-national actions that are tuned to local characteristics; regulatory and policy changes that encourage better learning and innovation across the society; matching budget with the broader policy targets; and be also focused not just employing horizontal policy measures.

2.3 Sustainability Transitions as an objective linked to innovation and industrial policy

Sustainability transitions are used in reference to wide-spread transformation of socio-technical systems that provide societal services, such as energy, food, transportation, and industrial systems more broadly (Köhler et al., 2019). Such transitions are characterised by directionality towards improved environmental and social sustainability, often described in terms of UN sustainable development goals. Visions, missions and targets are key elements of directionality in transitions (Geels et al., 2019). Constructing new directions can happen via processes of experimentation where new policy designs can help building new directions, networking where actors together create shared interpretations of future direction, and institutional change which can break past directions and make new ones (Kivimaa and Torrens, 2025).

In 2012, Weber and Rohrer noted the directionality failure of innovation policy – which may also partly apply to industrial policy (Weber and Rohrer, 2012). Subsequent policy efforts have tried to address directionality, for example, via the EU Green Deal, but it is not easy. Directionality efforts are complicated by **socio-technical multiplicity**, i.e. competing socio-technical configurations with their own actor-coalitions (Pel et al., 2020). For instance, in the energy sector, often contrasting positions are taken by the nuclear and renewable industry lobbies which both offer solutions for a zero-carbon energy system. In industrial policy, socio-technical multiplicity is linked to limited resources and having to pick selected industries to support.

Timing places a crucial role for policies, in terms of how to handle such multiplicity. Nurturing multiplicity is important, as it can be difficult for policymakers and even experts to forecast what becomes the “winning” solution in the long run. This is typically handled via innovation and commercialisation support (subsidies, low-interest loans, and even regulatory encouragement e.g. via emission-based rules). However, industrial support (e.g. subsidies for industrial clusters) need to be more selective in the world of limited resources. Such subsidies should be able to balance societal objectives pertaining to (future) competitiveness and employment, environmental protection and national security (e.g. Kivimaa et al., 2025). For example, if a choice needs to be made between two clusters that both hold potential for competitiveness but only one contributes to improved state of the environment, the choice should be the latter. It is important that industrial cluster support aims for building also future capabilities and is tuned to national or regional key competencies (e.g. around previous capabilities, geographical location and resources, social fabric etc.). For instance, not all EU regions can become future leaders in green hydrogen or batteries and other new industrial areas should also be sought.

Transitions are also characterised by **appraisal diversity**, indicating that the **level of ambition** for future direction and its **disruptiveness** are perceived differently by different actors. In industrial policy, this would refer to the degree of ambition and disruption to status quo to which, for instance, net-zero industries are pursued. Also, specific technologies, such as solar panels, can be used to form completely new systems or merely optimise existing systems (Yang et al., 2021). I argue that, **from a transitions perspective, modern industrial policy should not be closed for socio-technical diversity too soon, be ambitious and sufficiently disruptive too.**

Appraisal diversity is difficult to tackle via industrial policy as such, as all actors have their own perceptions. However, joint vision building exercises and the construction of industrial clusters can have as aims to bring different perspectives closer to each other, pertaining to the level of ambition. Disruptiveness of change brings with it also tensions – especially from those benefitting from status quo – but can be addressed by seeking opportunities to retrain employees to new industries and get incumbent fossil fuel industries to repurpose their assets (e.g. from offshore oil and gas to offshore wind power production).

In transition studies, the multi-level perspective is a frequently used analytical-conceptual framework. It describes how transitions happen in the interplay of new innovation niches accelerating and replacing existing systems, of existing socio-technical regimes (i.e. socio-technical systems) destabilising and shifting, and of external to the system landscape pressures influencing the process too (Geels, 2006). Industrial policy represents both the acceleration phase of transitions, when niches form new industrial sectors (e.g. cleantech) and the stability of existing socio-technical regimes, where attempts remain to maintain existing industries and preventing change (e.g. subsidies for energy intensive and fossil-fuel industries). I use the MLP frame to describe industrial policy measures in Section 4.

Mariana Mazzucato has argued that industrial policy should be challenge-focused and not sector-focused, where civil servants act as value co-creators and shapers and where public purpose is restored in policies to be aimed at creating benefits for citizens (Mazzucato, 2021). Mazzucato's thinking has given rise to mission-oriented innovation policy pursuits in academia (Janssen et al., 2021; Mazzucato, 2018a, 2018b), and in the European Commission and OECD alike. Instead, sustainability transitions research has given rise to thinking on transformative innovation policy (Borrás and Schwaag Serger, 2022; Kivimaa, 2022; Schot and Steinmueller, 2018) which is linked to mission-oriented innovation policy by sharing certain features, such as directionality, experimentation and inclusivity (Kivimaa et al. 2025). Mazzucato describes directionality in terms of "setting targets that are ambitious but also inspirational, able to catalyse innovation across multiple sectors and actors in the economy" by the public sector (Mazzucato, 2021, p. 6).

Both approaches to innovation policy emphasise experimentation as part of policymaking and governance of innovation and industrial development. Experimentation is seen as a way to create new ground and explore different alternatives in terms of direction, social and technical innovations, as well as institutional change (Kivimaa and Rogge, 2022; Sengers et al., 2019), providing the necessary reflexivity for governance in transitions. Examples of experimental approaches by the European Commission include, perhaps, the smart specialisation development and the setting up of EU missions.³ However, critique has been presented in terms of lacking ways to institutionalise experimentation as part of EU innovation policy (Foray et al., 2023).

The third aspect often emphasised in new kind of transformative innovation policy is inclusivity. Inclusivity means involvement of atypical actors into innovation (and industrial) policy processes, as well as being inclusive in creating benefits from innovation and industrial processes (see Boni et al., 2025). While EU just transition efforts have considered inclusivity in terms of cohesion between regions, other interpretations of inclusivity often appear lacking in EU innovation and industrial policymaking.

Most recent transitions literature has, on the one hand, emphasised acceleration and, on the other, the rather slow processes of decline in existing unsustainable socio-technical systems. This resonates with modern industrial policy in the sense that more support is needed for accelerating industrial transformation towards improved sustainability, competitiveness and resilience. On the other, we still face many barriers, as member states are keen to support established and incumbent industries, even when they struggle with some or all of these aspects. The Clean Industrial Deal is the European Commission's key attempt to tackle the challenges of climate change, reduced competitiveness of the EU and economic resilience. I will, hence, discuss this in the next section.

3. Clean Industrial Deal – synergies and tensions between the triple objectives

3.1 Clean Industrial Deal in the light of past industrial policy and sustainability transitions

The Clean Industrial Deal (CID) is Ursula Von Der Leyen's second commission's major policy initiative, representing a new growth and industrial policy strategy. It aims to address

³ " [T]he smart specialisation approach encourages experimentation on the policy level but the degree to which it favours experimentation by agents in regional economies depends on the case at hand." (Benner, 2019).

three challenges at the same time: “a climate crisis and its consequences, competitiveness concerns, and economic resilience”. To address the three challenges, or objectives, six elements are outlined: (1) affordable energy, (2) lead markets, (3) financing, (4) circularity and access to materials, (5) global markets and international partnerships and (6) skills. The Communication includes detailed policy measures for each element.

CID is not such a large shift from the Green Deal than what the name might imply. It does, however, remark a stronger shift to industrial policy and competitiveness as a potentially overriding policy objective over the objective of sustainability transitions. One such example is the reduced role of the Energy Efficiency First principle (Hermwille et al., 2025), despite it being clearly the clearest policy action bringing synergies between competitiveness, sustainability and security (Kivimaa et al. 2025).

In a certain way, CID returns to the post-war tradition of more selective and interventionist industrial policy. This is not just a development in the EU, but globally, where major states – especially the United States and China – are leading the way in setting tariffs and other conditions for trade with other countries.⁴ Yet, so far, at least the measures instigated under the Green Deal, such as NZIA, have not been too narrow – some even criticising the list of key technologies being too long (Kivimaa and Entsaló, 2025). Instead, others have criticised the lack of technology neutrality in the NZIA (Veugelers et al., 2024), related to government traditionally having been inefficient at picking winning solutions. However, Mazzucato has argued that the private sector may not be any better in picking winners than the public sector and that the arguments for the public sector not being able to pick winners is historically inaccurate (Mazzucato, 2021). She effectively argues that taking the mission economy approach, governments should pick a direction but within that direction take a broad portfolio approach. Essentially, this is something that NZIA has attempted to do. It is difficult to find a balance between a stronger industrial-interventionist approach relying on selecting certain technologies and sectors to support due to limited public resources and a more technology-neutral innovation-oriented approach. Both approaches have their shortcomings, and the earlier innovation-oriented industrial policy approach in the EU led to today’s challenges noted in the Draghi report.

Current limitations in public resources probably require taking a more selective approach in the European Commission and EU member states for the allocation of support for industries. However, when choices are made, it is important to keep plurality within EU industries (as noted above, not all regions can be leaders in the same technologies or industries) and not be dominated by larger member states’ favourites. Rather, what industrial policy should aim at is strengthening EU-based value chains across different industries and making the most of regional diversity (see Bertram et al., 2024).

Much of innovation and industrial development in the EU relies on cross-national innovation networks, funding arrangements and value chains – reaching also across the EU external borders. Hence, more inward-looking industrial policies may jeopardise innovation and industrial development in this sense, although if successful, the promotion of more EU-based innovation and industrial value chains would reduce external vulnerabilities (e.g. risks from foreign funding and ownership of strategic technologies and critical infrastructure, trade disruption risks caused by import-based value chains). Awareness of geopolitical risks has increased within the EU and the CID supports this, for instance, by resilience and security related public procurement criteria.

⁴ This development is against the initial idea of the World Trade Organisation to reduce trade barriers and promote free trade, although the EU has followed them more closely than countries, such as Taiwan and the United States (Chaisse and Su, 2025).

From a sustainability transitions perspective, the CID has retained the objective of decarbonisation but is almost quiet about other environmental problems (biodiversity loss, water pollution, etc.). The Communication on Clean Industrial Deal notes that “[i]ndustrial Decarbonisation Accelerator Act, will propose concrete measures to address permitting bottlenecks related to industrial access to energy and industrial decarbonisation, while maintaining environmental safeguards and protecting human health”, but it is not certain how this is made sure in the cleantech race. Some argue that how environment or social justice aspects (e.g. indigenous people’s rights) are accounted in the CRMA, depends on member state specific legislation – and may hence not be sufficient (Bogojević, 2024). Although, increasing critical raw materials extraction and processing in the EU balances out some of the larger environmental and social impacts caused by critical raw materials extraction in China and the Global South. Besides the critical raw material questions, the move towards joint procurement of natural gas will also risk creating new fossil fuel dependencies and reduce the climate ambitions of the CID (Hermwille et al., 2025).

3.2 Synergies and tensions between competitiveness, sustainability transitions and security

It is clear that no policymaking is without contradictions and tensions due to the nature of it in involving different societal objectives and differing interests in political negotiation. Nevertheless, it is important to be aware of potential synergies and trade-offs caused by specific policy objectives, policy instruments and policy mixes. Below, I assess, in two tables how industrial policy tools (Table 1) and the Affordable Energy Action Plan of the Clean Industrial Deal more specifically (Table 2) potentially affect the three objectives underlying the competitiveness compass, i.e. competitiveness, decarbonisation (while I take a broader lens of sustainability transition) and security (where I extend from economic resilience to other questions of security too). In the following, I will discuss some industrial policy instruments and their synergies and tensions between the objectives of competitiveness, sustainability transitions and security, briefly.

Tensions may occur, for instance, in association to **deregulation** aiming to boost industrial activity. For instance, if deregulation gives more opportunities again to use fossil fuels in industrial production to boost competitiveness, it undermines the sustainability and security objectives by causing increased greenhouse gas emissions and rising dependence on imported fossil fuels besides other negative environmental impacts. Therefore, it is vital that we have learned from past mistakes of massive environmental degradation and climate change caused by post-war industrial production and will not deregulate environmental protection (except in cases of overlapping bureaucratic arrangements). In cases, where environmental regulation is claimed to cause industrial decline, we should seek policy innovation instead of foregoing environmental quality standards.

Permitting times for renewable energy, electricity network and other related facilities have been noted to be excessively long in many member states (e.g. Draghi, 2024). Hence, making permitting faster is generally a policy measure to be supported. Here too, however, it is important to not overstep proper decision-making in areas of valuable nature (e.g. protected areas) or proper democratic and just processes where citizens are not only consulted but also heard. Nevertheless, there is much scope to reduce permitting times as currently the bureaucratic procedures involved are much too slow and inefficient. However, implementing the requirements of the CID, alongside of previous policies (RePower EU, NZIA, CRMA) is difficult as this requires more competent staff in public administrations, which are currently undergoing funding cuts in many places. A clear message is required

towards all the political parties that public sector capabilities are an important part of effective industrial policymaking.

Import tariffs are increasingly used in many countries across the world (e.g. Hufbauer and Jung, 2021) and can be beneficial in certain settings to boost domestic industrial production. The problem lies in potential or even likely counter-tariffs by concerned states, affecting their usefulness. From the perspective of sustainability transitions, tariffs can in the short-to-medium term slow down transformation, as technologies, components and other materials needed and produced outside Europe become more costly. In the case of the energy transition, for example, the EU is largely dependent on most solar power components and many critical raw materials from China. The inexpensive Chinese production has enabled a rapid acceleration of the energy transition to date (esp. solar power capacity). Nevertheless, without some protective measures on EU industries it may be impossible to keep industrial production and capabilities in Europe.

Promotion of industrial clusters is largely beneficial for all three objectives, if the European Commission and member states are able to correctly ‘pick the winners’ or if funding is granted following the principle of technology neutrality (i.e. different clusters can get funding as long as they contribute to decarbonisation or other sustainability efforts). Likewise, **investing in education and training** can be seen to be a synergy-promoting measure. **Industrial subsidies** can benefit competitiveness, sustainability transitions and security, if they are directed to sustainable industries based on EU-wide support. Member state specific industrial subsidies create unfavourable positions to smaller EU-member states with smaller resources for such subsidies.

In turn, **lowering energy taxation and tax exemptions** for energy intensive industries – a traditional industrial policy measure in industry-heavy countries – creates mostly negative effects on sustainability transitions by discouraging energy efficiency innovation and slowing down industrial transformation. Albeit applied to improve competitiveness by lowering energy costs, such cost-reductions can also be achieved by investments into electrification and energy efficiency. In the long term, such disincentives for improving energy efficiency may also negatively affect competitiveness of incumbent industry. Such measures also affect negatively on energy security by increasing consumption and associated supply risks from vulnerabilities of fossil fuel trade and supply chains and energy shortages. Instead, subsidies for the electrification of industries can create positive effects on transitions and security of supply – but involves risks caused by blackouts and technical disruptions in the electricity network. There are of course cases where high-energy using incumbent industries are vital for regional/national employment. Given many studies yet show the ineffectiveness of subsidies for energy-intensive sectors, the allocation of such subsidies should undergo improved assessments and scrutiny pertaining to their benefits and effectiveness.

Demand-side measures, such as public procurement or regulation (e.g. building codes) can benefit all three objectives, if designed right. Effective demand-side measures promote learning and innovation creation (aiming to bridge the innovation gap), sustainability transitions (aiming for sustainability) and security (when oriented to EU-based production). Similarly bilateral trade agreements may support all three objectives, if oriented to sustainability and lead to increased industrial production and competitiveness of the EU.

Table 1: Selected policy tools associated with industrial policy and their possible impacts on competitiveness, sustainability transitions and security

Policy tool	Competitiveness	Sustainability transitions	Security / Open Strategic Autonomy
General policy tools			
Deregulation in the domains of climate change mitigation and environmental protection	+ deregulation may level the playing field of European companies outside Europe, and within EU in the case of diverging national rules - deregulation may harm competitiveness by lost image of safe and environmentally friendlier products	+/- deregulation may have neutral effects on sustainability if same targets are maintained but bureaucracy reduced - deregulating may halt transition processes and increase pollution	- deregulating e.g. the energy sector may mean increasing fossil fuel imports creating security/OSA risks
Faster permitting	+ speeding up construction of new industrial facilities	+ speeding up renewable energy etc. climate-friendly projects - possible risks to nature protected areas and proper public participation	+ speeding up strategically important projects for security/OSA - fast permitting may jeopardise proper security risk assessments and create insecurity by bypassing democratic decision-making
Import tariffs	+ improves competitiveness of EU products/materials in the single market - reduces competitiveness on EU products relying on input of materials and components from third countries - may result in counter-tariffs, and losses of markets external to EU	+ creates opportunities to cleantech-industry in Europe - slows down transitions by stopping the diffusion of cheaper cleantech	+ improves security/OSA by increasing European manufacturing - possible counteractions may create shortages of critical materials or components
Promoting industrial clusters	+ improves competitiveness by creating more effective value chains - risks associated with 'picking winners'	+ creates industrial networks that may promote sustainability transitions - risks associated with 'picking winners' and environmental/social impacts	+ improves security/OSA by creating European networks - risks for international relations if disconnecting from international networks
Education and skills development	+ improves competitiveness by addressing skill and innovation shortages	+ improves sustainability by addressing skill and innovation shortages	+ improves security/OSA by addressing skill and innovation shortages
Industrial subsidies	+ improves competitiveness by levelling the playing field with countries external to EU - reduces competitiveness of smaller EU member states if conducted nationally - potential risk for fragmenting the single market with differing member state fiscal space	+ may support transitions if targeting cleantech and sustainability - harms transitions if addressed to incumbent unsustainable industries	+ improves security of supply /OSA by supporting EU industrial base
Lower energy taxation and exemptions for energy intensive industry	+ positive effects on competitiveness by lowering energy costs of industrial manufacturing - indirect negative effects by slowing down innovations and	- negative effect on sustainability transitions by discouraging energy efficiency and energy saving related innovations	- negative effect on security/OSA by increasing energy consumption and associated supply risks (e.g. import of fossil fuels, energy shortages)

	progress in energy efficiency and electrification	- slowing down industrial transformation and phasing out high-carbon activities	+ indirect positive effects by maintaining industries
Subsidies for electrification of industries	+ improves competitiveness if electricity reduces production costs in the long-term compared to natural gas	+ positive effect on transitions by moving away from fossil fuels	+ positive effect on security/OSA when associated with imported fossil fuels - risks associated with blackouts / technical faults
Demand-side measures (e.g. building codes, public procurement)	+ creates a domestic lead market, potentially leading to new competitive advantage via innovation diffusion - may not improve competitiveness of industries unless sufficient learning and innovative edge is obtained	+ creates opportunities for sustainability innovation development, diffusion and scaling up	+ may support security of supply / OSA if creating domestic alternatives to imported ones
Bilateral trade agreements	+ May create a demand-base for certain industries	+ May support transitions if involving cleantech/sustainability	+ supports OSA if encourages EU industrial production - risk to employment security if other country pulls out
Specific EU policies/laws			
Carbon border adjustment mechanism CBAM	+ aims to improve competitiveness in the EU - does not affect EU companies' competitiveness outside Europe - possible negative retaliation activities by states external to EU	+ aims to decarbonise industries in the EU and elsewhere	+ aims to level the playing field of European production vis-à-vis imported production, supporting OSA
Net-Zero Industry Act NZIA	+ aims to create competitive industries in the EU - similar measures in third countries might dilute the effects	+ aims to decarbonise EU industries - does not necessarily advance environmental sustainability in other aspects, might even worsen it	+ supporting keeping domestic industrial base in the EU - autonomy/security of new industrial base partly depends on where finance comes from
Critical Raw Materials Act CRMA	+ aims to increase critical raw materials production in the EU - large dependencies on critical raw materials remain despite the CRMA	+ supports decarbonisation of energy and transport + reduces environmental and social effects in the Global South, if European production replaces overall mining in third countries - creates new environmental effects, such as those from mining (water pollution, environmental degradation)	+ supports by reducing dependencies on third countries - large dependencies still remain, with vulnerabilities related to supply chains

Affordable Energy Action Plan (AEAP) includes eight actions: making electricity bills more affordable; bringing down the cost of electricity supply; ensuring well-functioning gas markets; energy efficiency delivering energy savings; completing the energy union; a tripartite contract to ensure affordable energy for Europe's industry; guarantee a security of supply for price stability, and; price crisis preparedness. The measures are not fully coherent regarding the different objectives of the Clean Industrial Deal – or the broader objectives of competitiveness, sustainability transitions and security (see Table 2). The eventual impacts of, for instance, the Affordable Energy Action Plan, depend on the preconditions set for each pillar and how they are implemented. It has been noted that fossil energy subsidies tend to hinder the diffusion of cleantech alternatives, while such alternatives are in the beginning more expensive than greenhouse gas intensive

production methods (Nilsson et al., 2021). It has also been noted that the support of natural gas imports creates new fossil fuel dependencies (Hermwille et al., 2025).

Table 2: Affordable Energy Action Plan – possible impacts on competitiveness, sustainability transitions, and security

	Competitiveness	Sustainability transition	Security
Making electricity bills more affordable Bring down cost of electricity supply A tripartite contract to ensure affordable energy for Europe's industry	Lowering energy costs for industry and encouraging electrification of industries (+) Lowering energy efficiency and demand reduction investments of industrial facilities if low electricity costs is reliant on subsidies (-) Boosting flexibility help reduce price spikes, also supported power by purchase agreements (+) Electricity transmission and distribution networks require substantial investments to be able to allow new industrial investments – this development may be hindered with aiming for lowering cost of electricity supply (-)	Encouraging electrification of industries (move away from gas) and enabling a just transition for consumers (+) May slow down new investments in energy efficiency, demand reduction and clean energy technology (-) unless support for such investments is part of this policy package (+) Boosting flexibility helps support renewable energy-based transition (+) Electricity networks require substantial investments to aid decarbonisation – this development may be hindered with aiming for lowering cost of electricity supply unless subsidised by governments (-)	Encouraging electrification of industries (move away from imported gas) (+) Depending on means, may slow down new investments in energy efficiency, demand reduction & domestic clean energy technologies (-) Boosting flexibility helps support security of supply in increasingly renewable energy-based systems (+) Electricity networks require substantial investments to enable security of supply – this development may be hindered with aiming for lowering cost of electricity supply (-)
Ensure well-functioning gas markets	Supporting incumbent gas-using industry (+) Possibly negative effects on cleantech industries replacing natural gas (-)	Discouraging electrification of industries and heating of buildings (-) Slowing down eventual phasing out of all fossil fuels (-)	Discouraging investments that support phasing out imported natural gas that creates vulnerabilities (-) Supports those parts of EU that still largely rely on gas for heating buildings and industrial production (+)
Energy efficiency delivering energy savings	Lowering costs for (energy-intensive) industries (+)	Lowering energy and resource use with positive effects on climate and nature (+)	Lowering imported energy and resources (+)
Completing the energy union	Increasing electricity interconnections supports electrification of industries (+) but may increase prices in certain member states (-)	Increasing electricity interconnections speeds up electrification of industrial and heating (+)	Increasing electricity interconnections improves overall energy security while contains risks of external damage (+)

The measures aiming to lower the cost of electricity supply and make electricity bills affordable are of course important for European competitiveness and keeping households out of energy poverty. The measures by which these objectives are pursued define their impacts on the three objectives of competitiveness, sustainability and security. If lowering costs are achieved by public subsidies, it is likely that such developments slow down investments in alternatives, such as energy efficiency, circular economy and electrification of heating and industries – which has negative implications on long-term sustainability transitions and energy security. The energy transition necessitates substantial investments

in new power transmission and distribution networks, which also need funding before such investments are completed. Related to this point, it has been argued that the CID has ignored the innovation and economic potential of circular economy (Hermwille et al., 2025). However, if the low costs are achieved by subsidies and other measures encouraging energy efficiency, renewable energy and electrification – which help bring down costs – AEAP will have more positive effects on the three objectives.

Support for gas markets is important in short-to-medium term, because in many EU member states, industries and building heating is dependent on gas. It is important to be aware that such policies at the same time slow down energy transition (i.e. decarbonisation) and maintain import dependencies. Instead, energy efficiency is a clear win-win-win strategy that should be kept high on the policy agenda. Likewise, completing the energy union is likely to support all three objectives, although may increase prices in those member states where lower electricity prices will increase by interconnectors to high-price states and regions.

4. Governance arrangements

In this section I briefly reflect on possible governance arrangements that can facilitate the implementation of the Clean Industrial Deal, noting connections to competitiveness, sustainability transitions and security. I will start by thinking about security – i.e. advancing the objectives of open strategic autonomy and technology sovereignty – and its links to sustainable transformation of industries. For this the OECD's categorisation of technology neutrality policies is useful. I will also illustrate this by making use of the multi-level perspective of sustainability transitions (see Figure 1).

The OECD (2023) has separated policies for technology sovereignty into *protection* (e.g., export control, foreign direct investment), *promotion* (e.g., bolstering industrial capacity, reducing reliance on foreign suppliers) and *projection* (e.g., STI alliances, technical standards). Such policies exist simultaneously, but the chosen policy mix may vary between countries due to diverse understandings of autonomy and sovereignty. These may also be seen as different stages of industrial support for a particular technology or industrial cluster, starting with projection-oriented activities supporting innovation, commercialisation and early diffusion, followed by promotion-oriented activities that support domestic industrial base, and then ending with protection-measures to support that industry. The projection, promotion and protection of new industries happens simultaneously as actions for protecting established industries and their potential steering towards decarbonisation (see elaboration in Figure 1).

In essence, the CID and the Energy Action Plan aim to both create and scale up new industries (green arrow) and maintain and decarbonise existing industries (blue arrow). Previous research shows that an essential part of transformative policy is also the vision and actions for the decline of old unsustainable industries, which can be more influential in terms of industry reactions than subsidies for innovation and scaling up (Kivimaa and Kern, 2016; Rogge and Johnstone, 2017). Hence, the overall modern transformative industrial policy mix needs to address the three processes (depicted below in different colour arrows).

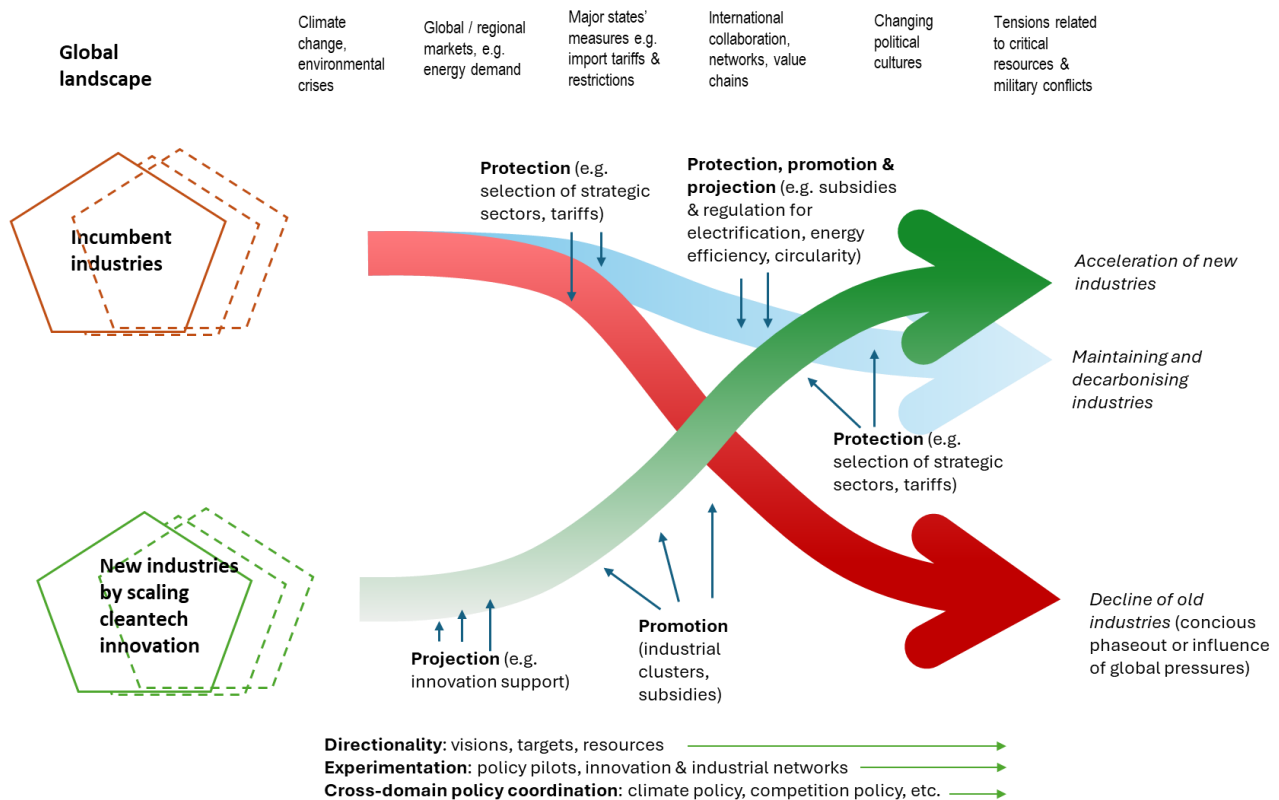


Figure 1: Timing and nature for industrial policy measures for new and established industries

The problem for governance and policy mixes lies with finding the optimal degree to which each activity is supported, as the effectiveness of the policy mix is ultimately also impacted by activities of other countries and multi-national companies, as well as its synergies and tensions with other policy domains (e.g. climate policy, security policy, foreign policy). Projection and promotion measures tend to be safer from the perspective of functioning international relations and are more likely to support sustainability transitions, but also suffer from competition. Other countries are equally likely to take up such measures, well seen in the global race to develop hydrogen and battery technologies.

On the EU-level, the selection of 'strategic' or 'critical' sectors for industrial policy to support within limited resources is difficult due to differing member state interests. For instance, the Net-Zero Industries Act has been criticised for including too many technologies (Kivimaa and Entsaló, 2025). Even before, the EU industrial policy has been argued to lack a vision (e.g. Bianchi and Labory, 2020). Limited financial resources and skills investments necessitate more **directionality** (see Section 2.3 and de Graaff et al., 2025). Directionality is also required by the objectives for sustainability and security. Hence, innovation-industrial policy should leave openness for innovation but also be able to make strategic choices where the EU holds competitive potential. For instance, industrial policy can find synergies between long-term competitiveness, sustainability transitions and security/strategic autonomy by promoting **energy efficiency, resource efficiency and circular economy -oriented industries**. They can be seen as win-win-win options.

Wider climate and energy targets also function as promoters of directionality. More broadly, there are long-term experiences in how environmental regulation -driven markets in Europe can generate competitive product and process innovations sold globally. As it will be difficult for European industries to be globally cost-competitive in any case,

industrial policy mixes must emphasise innovativeness and environmental sustainability, where new advanced technologies and social innovations may create new competitive advantages. Also the Draghi report recognised largest opportunities in the cleantech sectors (Draghi, 2024), even if China is already leading in the established solar power sector and has gained a strong position in batteries.

Veugelers et al. (2024) have called for **experimentation** as a key principle for green industrial policy and, for instance, regulatory sandboxes as frameworks for experimentation. Experimentation is likewise a key process of unfolding sustainability transitions (Kivimaa and Rogge, 2022; Sengers et al., 2019), contributing to reflexive and adaptive governance approaches along the development of transitions, often supported by intermediary organisations or platforms (Kivimaa, 2014; Kivimaa et al., 2019). The nature and role of experimentations needs to be defined more specifically for CID. The European Investment Bank (EIB) pilot programme for corporate Power Purchase Agreements (PPAs) mentioned as part of the CID is one kind of experimentation. Other types of experimentation should take place in innovation-industrial networks, coordination across policy domains, and also exploring the kinds of large **institutional changes** needed.

In the context of sustainable industrial transformation, Nilsson et al. have proposed a new innovation policy framework comprising: directionality, knowledge creation and innovation, **creating and reshaping markets, building capacity for governance and change**, international coherence, and **sensitivity to socio-economic implications of phase-outs** (Nilsson et al., 2021). Kivimaa et al. (2025) have identified similar and cross-cutting characteristics of challenge-led innovation policy, which can be applied to industrial policy too: directionality, experimentation and deep learning, **inclusivity and diversity**, regime destabilisation and phaseout, and comprehensive and adaptive policy mixes.

It is clear that in the present world, different policy sectors are increasingly intertwined and, therefore, **improved policy coordination** across domains and levels of administration and policy coherence are needed. The European Commission has certain measures in place to facilitate coordination and coherence, such as the multiannual framework and interservice consultations – and the Competitiveness Compass and CID provide grounds for improved coordination. However, research has noted gaps between the mandates and activities of different DGs, and much more coordination and coherence are needed to tackle the persistent challenges we face.

Another challenge is the increasing knowledge and skills needs required from public service to coordinate and implement multi-objective industrial policy – while in many member states public organisations have been subject to budget cuts. It is important to keep and increase knowledge capacity in public organisations – without excessive reliance on consultants – to be able to sufficiently manage modern industrial policy. Table 3 summarises some of the key aspects for governing modern industrial policy mentioned above – and ways to implement them.

Table 3: Examples of governance arrangements for modern industrial policy

Dimension of governance	Examples of ways to implement
Directionality	• Setting aspirational targets • Building and integrating industrial clusters for win-win(-win) solutions, especially around circular economy, energy efficiency, cleantech and dual use • Improved coordination/coherence between industrial, trade, finance, energy, transport, climate, environmental, agri-food, foreign and security policies • Assessing when regulations can create/expand niche markets that hold potential for upscaling, e.g. around technologies/services for electrification, building technologies, green-digital innovations • Continue RTD subsidies that support multiplicity but be more selective with industrial subsidies (decide foci areas but should be differentiated for different regions and aiming for EU-based value chains) • Conducting joint vision building exercises aiming to reduce appraisal diversity but being inclusive • Improving (regional) directionality of smart specialisation and correct previous shortcomings
Experimentation and deep learning	• Creating spaces to learn new things and unlearn established practices and ways of thinking • Allowing failure, discuss learnings from failure openly • Pooling experiences from multiple experiments (regulatory sandboxes, policy labs, living labs, pilot programmes, etc.) – institutionalise successful examples (i.e. moving from temporary status/funding to permanent status/funding for initiatives) • Combining experimental approaches with supportive institutional changes (e.g. regulatory changes supporting the experiments, or organisational changes supporting new ways of doing and organising) • Evaluating experimentation and experiments when they are not useful for learning or innovation • Investments into education and training
Creating and reshaping markets	• Using public procurement criteria and other demand-side measures to support win-win solutions • Faster permitting by removing unnecessary bureaucratic hurdles and improving public sector resources and capabilities (see below) • Assessing when regulations can create/expand niche markets that hold potential for upscaling, e.g. around technologies/services for electrification, building technologies, green-digital innovations (<i>same as under directionality</i>) • Allowing temporary relaxations to single market rules to allow new market creation in regions (based on proper assessments) • Selective use of import tariffs
Managing phase-out of fossil fuels	• Clean-tech market creation with foreseeable development of phase-down and phase-out policies, supported with policies such as ETS and CBAM • Supporting reorientation of businesses and retraining of employees to new (cleantech and dual use) sectors • Supporting electrification and energy efficiency of high-energy industries to lower costs of operation
Strengthening public sector capabilities	• Reserving sufficient resources for public servants to be innovative, reflexive and anticipatory of developments in technologies, markets, environment, other connections • Encouraging interactions across departments and governance levels • Reducing the role of consultants in vital knowledge-sensitive areas

5. Concluding remarks

In the concluding section, I reflect on the CID from the perspectives of the three objectives (competitiveness, sustainability and security) and outline some key ingredients for modern transformative innovation policy.

It is clear that the current challenges of climate change, the EU's reduced competitiveness and mounting insecurity, require **well-targeted and directional industrial policy**. The objectives of the Clean Industrial Deal are a step towards the right direction, and when implementing it, it is vital that competitiveness pursuits will not sacrifice the climate and environmental objectives of the EU. Actions for the latter, in turn, are paramount for long-term survival of the planet, economic resilience and security in the decades to come. In other words, there is no economic resilience and societal security without ambitious efforts to decarbonise, protect our natural environment and adapt to the impacts of climate change.

Directionality in modern industrial policy is essential. This is required to set the way to future industrial, economic and societal wellbeing in Europe. An approach to directionality, should, first, **leave space for openness of collaboration and innovation** (akin to open strategic autonomy that does not forget openness), second, **invest in win-win-win options** (especially energy efficiency/sufficiency and circular economy) and, third, **set visions, targets and regulations that guide the way**. Public subsidies and taxation also play an important role in directionality, but research has well demonstrated the importance of regulation in new market formation. Such regulation should include measures to phaseout unsustainable industries in the long-term (e.g. fossil fuels, animal fur production) providing an enforced vision for the future. Stability is needed in terms of directionality but not in terms of maintaining status quo. This means **anticipatory and gradually strengthening policy development**. Companies have for decades as well as today called for consistent and foreseeable policy development, which can be supported with improved directionality.

In a time of limited resources, choices need to be made. Here I support Mazzucato's portfolio approach, where broader visions are set and a range of (selected) industrial clusters supported within those visions. An important part here is to **strengthen public sector capabilities** to actively take part in vision and portfolio setting. **Experimentation and reflexive governance are also important**. We may not get the vision right on one go. Surrounding geopolitical, environmental and health crises may change the operating conditions, which means policy instruments need to change along the way. Experimentation and learning are also important to find synergies between the different policy objectives in terms of concrete policies and social and technical solutions setting ground for future industries.

The discussion in this paper showed that industrial policy measures with most synergies between competitiveness, sustainability transitions and security include **promotion of industrial clusters, investments into education and training, and EU-wide industrial subsidies on sustainable industrial production**. **Faster permitting is also something that needs to be pursued, by removing unnecessary bureaucratic hurdles, but not by omitting thorough environmental impact assessments and social justice considerations**. In contrast, industrial policy measures that involve tensions and should be avoided (where alternative policy measures can be found) are deregulation pertaining to environmental protection, high import tariffs, and subsidies for energy-intensive industry. Pertaining to energy industry measures, the objectives for lowering energy supply costs

and price to consumers, need to be achieved by subsidies for advancing energy efficiency and clean energy solutions which lower the costs, and not by subsidising energy-intensive activities. However, independent of energy source, low-income households need support to avoid the energy poverty trap.

Innovation, decarbonisation of industries and new industries built around sustainability solutions (e.g. renewable energy, batteries, circular economy) should be at the core of modern industrial policy. Questions remain how wide the selected pathways should be (ranging from regional industrial clusters to larger directional portfolios). The European Commission has done extensive work in regional innovation policy and supporting interactions between regions, which should be continued, to be of use of broader organisation of EU industries (for instance, in vision building processes). The unique strength of diverse EU regions has been noted elsewhere to have been neglected (Bertram et al., 2024). EU-level coordination of industrial policy (visions, regulation, subsidies) is vital to place member states in similar settings and a level playing field in terms of competitiveness, sustainability and security. I argue that, from a transitions perspective, modern industrial policy should not be closed for socio-technical diversity too soon, be ambitious and transformative, and sufficiently disruptive too.

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